

Unit 4

1. In a 4×4 array multiplier, how many Full Adders (FA) and AND gates are required?

- A. 16 AND + 12 FA
- B. 16 AND + 8 FA
- C. 12 AND + 16 FA
- D. 8 AND + 12 FA

Answer: A $\rightarrow (n^2 = 16 \text{ AND gates}, FA = n \times (n - 1) = 12)$

2. For an $n \times n$ array multiplier using **AND** and **Full Adder** cells, each FA has 2 gate delays and each AND gate has 1 gate delay.

What is the **worst-case propagation delay** (in gate delays)?

- A. $2n + 1$
- B. $4(n - 2) + 2n + 1$
- C. $6(n - 1) - 1$
- D. $n^2 + 1$

Answer: C $\rightarrow \text{Delay} = 6(n - 1) - 1$

3. For two unsigned n -bit numbers multiplied using array multiplication, the result has —

- A. n bits
- B. $n + 1$ bits
- C. $2n$ bits
- D. n^2 bits

Answer: C $\rightarrow \text{Product of two } n\text{-bit numbers} = 2n\text{-bit result}$

4. In a 4×4 array multiplier, if one AND gate delay = 1 ns and one FA delay = 2 ns, estimate the worst-case delay.

- A. 20ns
- B. 18 ns
- C. 17 ns
- D. 25 ns

Answer: C $\rightarrow \text{Delay} = 6(n - 1) - 1 = 17 \text{ gate delays} \approx 17 \text{ ns}$

5. In Booth's multiplication algorithm, when the pair of bits $(Q_0, Q_{-1}) = (1, 0)$, the operation performed on the accumulator A is:

- A. Add multiplicand (M)
- B. Subtract multiplicand (M)
- C. No operation (shift only)
- D. Add twice multiplicand ($M \times 2$)

Answer: B — *Booth rule: (1 0) $\rightarrow -1 \times M \rightarrow$ Subtract multiplicand.*

6. Bit-pair recoding in multiplication reduces:
- A. Number of partial products
 - B. Hardware complexity
 - C. Number of control signals
 - D. Both A and B

Answer: D — *Bit-pair recoding halves the number of partial products and simplifies control logic.*

7. If three 4-bit numbers are added using a CSA, what is the bit-width of the carry output after alignment (left shift)?
- A. 4 bits
 - B. 5 bits
 - C. 8 bits
 - D. 3 bits

Answer: B. 5 bits

8. For $k = 16$ summands, the number of CSA levels needed to reduce them to two vectors
- A. 3.4
 - B. 4.1
 - C. 5.1
 - D. 6

Answer: D 6

9. In the restoring division algorithm, if after subtraction the sign bit of register A is 1, then which of the following operations is performed next?
- A. Left shift Q to A by one position
 - B. Subtract M again from A
 - C. Add M to A (restore) and set $q_0 = 0$
 - D. Add M to A (restore) and set $q_0 = 1$

Answer: C

Explanation: When the remainder becomes negative (sign bit = 1), restoration is performed by adding back the divisor ($A = A + M$) and the quotient bit $q_0 = 0$

10. What is the value of accumulator and register Q at end of first cycle for restoring division of 1110/0011.

- a. Accumulator= 00001 Q=1100
- b. Accumulator= 11001 Q=1101
- c. Accumulator= 00100 Q=1100
- d. Accumulator= 00001 Q=1001

ans: a

11. The Value of accumulator and carry bit at end of 2nd cycle for multiplication of 2 numbers 12(M) and 15(Q) using shift add multiplication method is: (2marks)

- a. Accumulator= 1001 Carry bit=0
- b. Accumulator =1001 carry bit=1
- c. Accumulator =0010 carry bit=1
- d. Accumulator=1111 carry bit=0

ans: a

12. In both restoring and non-restoring division, the quotient bit (q_0) depends on:

- A. The sign of M
- B. The sign of A
- C. The sign of Q
- D. Carry from adder

Answer: B

Explanation: Quotient bit is determined by the sign bit of A after each operation.

13. Given the following binary number in 32-bit (single precision) IEEE-754 format:

00111110011011010000000000000000

The decimal value closest to this floating-point number is

- a. 2.3×10^{-1}
- b. 3.3×10^{-1}
- c. 4.3×10^{-1}
- d. 5.3×10^{-1}

Ans:

Exponent bits = E = 01111100 = 124 (in decimal)

Mantissa bits = M = 110110100000000000000000

In IEEE-754 format, 32-bit (single precision)

$$\begin{aligned}
 & (-1)^s \times 1.M \times 2^{E-127} \\
 &= (-1)^0 \times 1.1101101 \times 2^{124-127} \\
 &= 1.1101101 \times 2^{-3} \\
 &= (1 + 2^{-1} + 2^{-2} + 2^{-4} + 2^{-5} + 2^{-7}) \\
 &= 0.231 = 2.31 \times 10^{-1} \approx 2.27 \times 10^{-1}
 \end{aligned}$$

14. In IEEE floating point representation, the hexadecimal number 0xC0000000 corresponds to

- 1. -3.0
- 2. -1.0
- 3. -4.0
- 4. -2.0

Ans: Ans:

Exponent bits = E = 10000000 = 128 (in decimal)

Mantissa bits = M = 000000000000000000000000

In IEEE-754 format, 32-bit (single precision)

$$\begin{aligned} & (-1)^s \times 1.M \times 2^{E-127} \\ & = (-1)^1 \times 1.0 \times 2^{128-127} \\ & = -1 \times 1.0 \times 2 \\ & = -2.0 \end{aligned}$$

15. During a memory read sequence using MAR and MDR, which control signals are correctly asserted first?

- A. MARin then Read
- B. Read then MARin
- C. MDRin then MARin
- D. MDROUT then MARin

Answer: A

Explanation: The address must be loaded into MAR before issuing the Read command.

16. What is the role of a tri-state gate in register organization?

- A. Amplifies the bus signal.
- B. Acts as a decoder for address selection.
- C. Connects or isolates register output from the common bus.
- D. Generates clock pulses for register transfer.

Answer: C

Explanation: The tri-state buffer enables or disconnects the register from the bus without interfering with other devices.

17. When the control signal Riin = 1 for a register, what operation is performed?

- A. The register places its contents on the bus.
- B. The bus data is loaded into the register.
- C. The register output enters high-impedance state.
- D. The register clears its contents.

Answer: B

Explanation: Riin = 1 activates the input control, causing the D-flip-flops of the register to load bus data on the next clock edge.

18. During register transfer R1 → R2, which control signal sequence is correct?

- A. R1in=1, R2out=1
- B. R1out=1, R2in=1
- C. R1out=1, R2out=1
- D. R1in=1, R2in=1

Answer: B

Explanation: R1out enables the source register onto the bus; R2in loads bus data into destination register

19. In a branch instruction, the Program Counter (PC) is replaced by the _____.

- A. Instruction Register (IR) contents

- B. Branch Target Address
- C. Condition Code Register value
- D. Offset value X only

Answer: B

Explanation: A branch instruction replaces the PC contents with the branch target address, which is obtained by adding the offset X to the next sequential address.

20. Which of the following control sequence corresponds to a conditional branch operation?

- A. Offset-field-of-IRout, Add, Zin, If N = 0 then End
- B. Rout, MARin, Read, WMFC
- C. Yin, Add, Zout, Rin
- D. PCout, MARin, Read

Answer: A

Explanation: This sequence computes the branch target and loads it into Z only if the condition code is true.

21. Which of the following register sequences is correct for fetching an instruction?

- A. PCout → MARin, Read, WMFC, MDRout → IRin
- B. IRout → PCin, MDRout → MARin
- C. MDRin → PCout → IRin
- D. Zout → Yin → MARin

Answer: A

Explanation: This represents the typical fetch sequence used in single-cycle datapaths.

22. What is the role of the Incrementer unit in the processor datapath?

- A. To add two register values
- B. To add 4 to the PC without using the main ALU
- C. To store ALU results temporarily
- D. To control the tri-state bus signals

Answer: B *Explanation:* The Incrementer bypasses the main ALU for PC increment operations, reducing load and latency

23. In a hardwired control unit, the control signals for each instruction are generated based on:

- A. Microprograms stored in ROM
- B. Control step counter, instruction register, and condition flags
- C. Address decoding logic in the MAR
- D. A software loop maintained in main memory

Answer: B

Explanation: Hardwired control uses combinational logic driven by the control-step counter, instruction decoder, and condition codes

24. Which of the following statements about hardwired control is correct?

- A. It is flexible and easy to modify for new instructions.
- B. It executes microinstructions sequentially.
- C. It is faster but less flexible than microprogrammed control

D. It stores microinstructions in control memory.

Answer: C, Explanation: Hardwired control achieves high speed through fixed logic but is difficult to modify for complex or evolving instruction sets

25. A **Control Word (CW)** in a microprogrammed unit:

- A. Is an instruction for the ALU
- B. Represents a set of simultaneous control signals
- C. Contains only address bits of data memory
- D. Holds flag register values

Answer: B

Explanation: Each CW specifies which control lines (e.g., PCout, Yin, Add) are active during a micro-operation

26. The **microprogram counter (uPC)** in a microprogrammed control unit increments with each microinstruction except when:

- A. $MFC = 0$
- B. A Branch μ -instruction or End μ -instruction occurs
- C. $RUN = 1$
- D. $WMFC = 1$

Answer: B

Explanation: For branch or end microinstructions, the uPC loads a new or starting address instead of incrementing sequentially